Besides these, BT4 comes with other new and notable features:

- Full functional support for Pico E-12 and E-16 cards, which can be used to accelerate WPA PSK and Bluetooth cracking with tools such as coWPAtty.
- Support for building and executing CUDA- (Compute Unified Device Architecture) powered applications, which exploit the computing power of graphics processing hardware (GPUs) from NVIDIA. Two important CUDA-based tools available in BT4 are CUDA-Multiforcer and Pyrit. CUDA-Multiforcer can be used to accelerate password cracking, whereas Pyrit helps in breaking WPA/WPA2 PSK Wi-Fi networks faster.
- BT4 supports PXE cards, and hence can be booted over the network.
- Inclusion of the popular vulnerability assessment tool, SAINTexploit.
- Inclusion of the latest version of the popular information-mining/gathering tool, MALTEGO 2.0.2.
- Latest 802.11 wireless injection patches are applied for comprehensive support of wireless injection.
- Availability of tools for RFID auditing.
- Enhanced functionality of the popular information-gathering and security-auditing tool, Unicornscan.

BackTrack installation

BackTrack is most popular as a live DVD (the ISO image is over 1.5 GB) or a live USB; there is no need, as such, to permanently install it on a particular host. To create a live USB, use UNetbootin (*http://unetbootin.sourceforge.net/*) on either Windows or Linux, give it the path to the BackTrack ISO image, and select the target USB drive…Simple.

However, since BT4 is based on Ubuntu, and has many general applications as well, a permanent installation to hard disk could be useful for the frequent use of its security tools as well as the general applications. To install, boot BT4 from the live DVD/USB, and then run *startx* at the command line to bring up the GUI. Click the *install.sh* script on the desktop to begin installing BT. After selecting the time zone and keyboard settings, the partition configuration screen offers you options for: automatic partition configuration, taking into account already-occupied space on the disk; installing BT on your entire disk; and manually configuring partitions. After partitioning is done, accept the installation summary, and click 'Install'. In a few minutes, installation is done; GRUB will be configured automatically, if you have two or more operating systems installed on the hard disk (including BT).

BackTrack can also run in a virtual machine (for example, VMware Player running on a Windows host). You can either install into a new VM, with the BT ISO as the source, or download a pre-built VMware image from the website.

BackTrack usage

The default BT username and password is *root/toor*. After BT is booted, the network interfaces are disabled by default, for security reasons. You have to bring them up explicitly, using the *ifconfig* command. If you want to acquire an IP address for an interface from a DHCP server on the network, use the *dhclient* command.

As mentioned earlier, BT4 boots to a command line, and you need to run *startx* explicitly to bring up the GUI. BackTrack has a simple KDE-based desktop by default, running a lightweight KDE3 window manager. The desktop provides easy navigation to security tools and other useful applications. Since most of the security tools are command-line utilities, clicking on such a menu item opens a new shell and displays the help message associated with the command. An interesting element called the *Run* box is embedded in the lower panel, which allows you to run applications without starting a terminal application.

BT4 contains a variety of open source security tools for various purposes. These are available in the following main categories:

- Information gathering
- Network mapping
- Vulnerability identification
- Penetration
- Privilege escalation
- Maintaining access
- Radio network analysis
- VoIP analysis
- Digital forensics
- Reverse engineering

Apart from the security tools, the BT4 distribution also includes Internet applications like Firefox, and various chat clients like Skype. Also, various services such as HTTP, FTP, VNC, Remote Desktop and TFTP are available; these can be easily started and stopped using the navigation menu on the desktop.

BT4, with the richest collection of security tools and other useful applications, has become today's most popular self-sustained Linux distribution. It's available for free, and is very easy to use. The distribution can be used for comprehensive security assessment of wired and wireless networks, Web applications, network servers and computer programs. The tool-kit can also be useful in learning network behaviour, troubleshooting network problems, digital forensic investigations and development testing. One can also customise the BackTrack distribution with other tools, and use it for many purposes, such as demos, marketing and training. The popularity of BT4 can be gauged from the fact that the distribution has already witnessed over a million downloads since its release. END

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